

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Open-label Study to Evaluate the Safety and Efficacy of MK-3120 in Participants With Advanced Solid Tumors **Short Title/Acronym:** MK-3120-002 **Protocol Number:** MK-3120-002 **NCT Number:** NCT06818643 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-3120 (also known as SKB410 / SKB-410, an investigational Nectin-4 targeted antibody-drug conjugate or ADC) **Phase of Development:** Phase I/II **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of MK-3120 and to determine the recommended Phase 2 dose/dosing schedule in participants with advanced solid tumors. **Secondary Objectives:** To evaluate the preliminary efficacy of MK-3120 by assessing the Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS) as assessed by the investigator per RECIST 1.1 criteria. Additional secondary objectives include characterizing the pharmacokinetics (Cmax, Cmin, AUC) of the MK-3120 ADC, total antibody (TAb), and free payload.

2.2 Background & Rationale Researchers are investigating new approaches to treat patients with certain advanced solid tumors—cancers that have spread to other body parts and cannot be surgically removed. As current treatment options are limited for patients whose cancer has progressed on standard therapies, the primary purpose of this study is to gather data on the safety profile of MK-3120, assess patient tolerance across different dose levels, and evaluate its potential to halt or reverse tumor progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Dose-finding (Multiple experimental dose levels) **Blinding:** Open-label (None) **Randomization:** Randomized (Parallel Assignment to Dose Level 1 vs. Dose Level 2)

3.2 Planned Enrollment & Duration Target \$N\$: 270 participants **Number of Sites:** International study spanning across 42 countries **Estimated Study Start:** 25-Mar-2025 **Estimated Primary Completion:** 25-Oct-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Has a confirmed advanced (unresectable and/or metastatic) solid tumor and has received or been intolerant to all available standard treatments. 3. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 3 days before intervention allocation/randomization, along with adequate organ function. 4. If HIV positive, must have well-controlled HIV on antiretroviral therapy (ART); if HBV or HCV positive, must have undetectable viral load and be on appropriate antiviral therapy.

4.2 Exclusion Criteria 1. Known additional malignancy that is progressing or has required active treatment within the past 2 years, or known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 2. History of active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 3. History of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 4. Has symptomatic pleural effusion, ascites, and/or pericardial effusion requiring repeated drainage, or uncontrolled cardiovascular/cerebrovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-3120 administered at experimental Dose Level 1 or Dose Level 2 as per the randomized schedule specified in the study arm. Route of Administration: Intravenous (IV) Infusion Duration of Treatment: Continues until documented disease progression (PD), unacceptable toxicity, or study withdrawal (evaluated up to approximately 72 months).

5.2 Concomitant & Prohibited Therapy Permitted: Antiretroviral therapy (ART) for HIV-positive participants and antiviral therapy for HBV/HCV; standard supportive care and symptom management. Prohibited: Concurrent investigational agents, systemic immunosuppressive medications for inflammatory bowel disease, and therapies strictly excluded per oncology trial guidelines (e.g., live vaccines).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Imaging (RECIST 1.1), Safety Labs
Baseline	Day 0	Randomization, First Dose (IV Infusion), PK Sampling Initialization
Interim	Per Cycle	Safety Labs, PK Sampling (Cmax, Cmin, AUC), Efficacy Assessment via Tumor

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and Tolerability: Evaluated by the number of participants who experience at least one Adverse Event (AE) and the number of participants who discontinue study treatment due to an AE. **Measurement Method:** Clinical safety assessments and AE reporting per standard regulatory criteria.

7.2 Secondary & Exploratory Endpoints * **Objective Response Rate (ORR):** Percentage of participants achieving Complete Response (CR) or Partial Response (PR) per RECIST 1.1. * **Duration of Response (DOR):** Time from first documented CR or PR until disease progression or death. * **Progression-Free Survival (PFS):** Time from first dose to documented PD or death due to any cause. * **Overall Survival (OS):** Time from first dose of study treatment to death from any cause. * **Pharmacokinetics:** Area under the curve (AUC), Cmax, and Cmin of MK-3120 ADC, total antibody, and free payload.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Monitored continuously. An AE is defined as any untoward medical occurrence temporally associated with the use of the study treatment, whether or not considered related. **Serious Adverse Events (SAEs):** Standard expedited reporting timeline (typically 24 hours) for all life-threatening events, hospitalizations, or fatalities. **Data Monitoring Committee (DMC):** Internal safety review committee coordinated by the Sponsor to assess dose-limiting toxicities during the dose-finding phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * **Safety Set:** All participants receiving at least one dose of MK-3120. * **Intent-to-Treat (ITT):** All randomized participants used for evaluating efficacy measures (ORR, DOR, PFS, OS). **Sample Size Justification:** N=270 participants across parallel assignment groups to provide sufficient statistical power for identifying dose-limiting toxicities and establishing preliminary efficacy in advanced solid tumors. **Handling of Missing Data:** Right-censoring will be used for time-to-event outcomes (e.g., PFS, DOR, OS) for patients without events at the time of data cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard phase 1/2 risk-based monitoring plan managed globally across participating trial sites. **Audit Trail:** Secure Electronic Data Capture (EDC) system with full audit trails for eCRF lock and data clarification query resolutions.

11. Study Identifiers & Governance

Primary ID: MK-3120-002 **Registry Number:** NCT06818643 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study to Evaluate the Safety and Efficacy of MK-3120 in Participants With Advanced Solid Tumors (MK-3120-002) **Official Regulatory Title:** A Phase 1/2 Open-label Study to Evaluate the Safety and Efficacy of MK-3120 in Participants With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced Solid Tumors (Malignant Neoplasms)

Diagnosis Threshold: Confirmed advanced (unresectable and/or metastatic) solid tumor after failure of standard available treatments

Medical Methodology: Investigational Nectin-4 Antibody-Drug Conjugate (MK-3120) **Optimization Target:** Recommended Phase 2 dose (RP2D) identification, ORR, and Tolerability

13. Study Procedures & Methodology

Management Pathway: Standard Oncology Phase 1/2 Clinical Trial Pathway

Education Protocol (HCP): Investigator Brochure review and Site Initiation Visits

Education Protocol (Patient): Informed consent outlining trial risks/benefits, and detailed visit schedule guidance

Quality Audit Mechanism: Source Data Verification (SDV) by Clinical Research Associates (CRAs)

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 72 months

Onsite Visit Cadence: Regularly scheduled IV infusions and RECIST 1.1 tumor imaging assessments

Remote Monitoring Frequency: Continuous AE tracking and survival follow-up **Checklist Review Interval:** Prior to each dosing cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					